OFFSHORING POLLUTION

To save the state's polluted rivers, the Gujarat government plans to dispose of the bulk of industrial wastewater directly into the deep sea. But environmentalists fear the move's impact on marine life and coastal settlements in the Gulf of Khambhat

By Jumana Shah



If all goes as planned, Gujarat over the next decade will have an extensive network of deep-sea pipelines to discharge effluents from its seven industrial clusters—Ahmedabad, Vadodara, Ankleshwar, Surat, Vapi, Sarigam and Jetpur—into the Arabian Sea. Aiming to save the Sabarmati and other critically polluted rivers of the state, the Gujarat government informed the Vidhan Sabha in September that seven such deep-sea pipelines, also known as marine outfalls, are in various stages of planning and execution. The state's first such major pipeline was made operational in 2016 in Bharuch district—from Jhagadia industrial estate to the coastal village of Kantiyajal about 60 km away—which disperses treated effluents into the Gulf of Khambhat at a depth of 11 metres, 9.4 km from the shore.

While the Rs 8,000 crore-plus project is expected to give rivers flowing through the state a new lease of life, environmentalists have raised concern over its possible impact on marine ecosystems and coastal habitations. Industries in Gujarat produce 575 million litres per day (MLD) of effluents. Of this, 60 MLD, or 11 per cent, is discharged directly into the sea, and the remaining 515 MLD, or 89 per cent, is let into various rivers and creeks.

The Central Pollution Control Board (CPCB) in 2018 identified 20 polluted river stretches in Gujarat, of which five—Sabarmati, Bhadar, Bhogavo, Amlakhadi and Vishwamitri—are classified as 'critically polluted'. The pollution in the Sabarmati in Ahmedabad and Mahi near Vadodara has, in particular, become unbearable due to its detrimental impact on aquatic life and the lingering stench. Untreated and illegal discharge of industrial wastewater and domestic sewage into these non-perennial rivers has been identified as the root cause.

THE NEED FOR OUTFALLS

According to the Comprehensive Environment Pollution Index (CEPI), the industrial clusters of Vatva and Narol in Ahmedabad, Ankleshwar in central Gujarat and Vapi in south Gujarat have been among the 'critically polluted areas' since 2018—leading to a CPCB moratorium on any expansion in these clusters. Looking for a solution at the time, representatives of multiple industrial estates had urged the government to construct a network of marine outfalls in the state. This is because the norms for effluent discharge into the deep sea are relaxed in comparison to those for rivers, thus reducing the cost of treatment.

Take, for instance, the permissible limit for biochemical oxygen demand (BOD) and chemical oxygen demand (COD)—both used to determine the amount of organic pollution in liquid waste. While the CPCB mandates a maximum BOD of 30 milligrams per litre and COD of 250 mg per litre in the case of treated effluents being discharged into rivers, the norms have been relaxed to 100 mg and 500 mg, respectively, when it comes to deep-sea discharge. Why such relaxation? Because marine outfalls rely on the assimilative capacity of the sea to further dilute any detrimental effects of industrial waste.

"By implementing deep-sea disposal solutions and providing ap-